

Curriculum Vitae

Affiliation

Name	Prof. Dr. Cornelia Schneider (Interim Professor)
Address	FAU Erlangen
	Mathematics and Data Science Departments
	Cauerstr. 11
	91058 Erlangen, Germany
Phone	+49-(0)9131-8567207
Email	cornelia.schneider@fau.de
Homepage	https://corneliaschneider.github.io/

Education and Professional Career

2025–2027 **Substitute W3-professorship "Data Science"**, *FAU Erlangen, Germany*

07/2020 **Habilitation**, *Philipps-University Marburg, Germany*
Habilitation thesis "Besov regularity of partial differential equations, and traces in function spaces"

since 2018 **Akademische Oberrätin (Assistant Professor)**, *FAU Erlangen, Germany*
2023–2024 parental leave

2010–2018 **Akademische Rätin (Lecturer)**, *FAU Erlangen, Germany*
2013–2014 parental leave

2009–2010 **Postdoc**, *Universidade de Coimbra, Portugal*

2006-2009 **PhD**, *University of Leipzig, Germany*, Junior research team "Analysis, geometry and their interaction with the natural sciences"
PhD thesis "Besov spaces with positive smoothness", supervisors: Prof. D. D. Haroske and Prof. T. Kühn

2004–2006 **Graduate Studies**, *FSU Jena, Germany*, Mathematics with minor Physics, Diplom, Grade: 1.0 (passed with distinction)
Master thesis "Growth Envelopes in Sobolev Spaces", supervisor: Prof. D. D. Haroske

2003–2004 **Study abroad**, *University of South Australia in Adelaide, Australia – University of Auckland, New Zealand*

2000–2002 **Undergraduate Studies**, *FSU Jena, Germany*, Mathematics with minor Physics

1992–2000 **School**, *St. Josef Gymnasium Dingelstädt, Germany*, A-level, Grade: 1.0

Grants

2024 **Grant of the German Research Foundation (DFG)**, *Germany*, for International Scientific Events/Annual Conferences (20.000 Euros)
Together with D.D. Haroske.

2020-2021 **Grant of the German Research Foundation (DFG)**, *FAU Erlangen-Nuremberg, Germany*, temporary position for principal investigators (139.600 Euros)

2017-2019 **Grant of the German Research Foundation (DFG)**, *FAU Erlangen-Nuremberg, Germany*, temporary position for principal investigators (177.000 Euros)

2010 **Grant of the Foundation for Science and Technology (FCT) (3+3 years)**, *Portugal*, declined due to permanent position in Erlangen

Scholarships

2024 **DAAD scholarship and travel expenses**, *Germany*
3 months visit to Argentina for my PhD student Nick Schneider

2023 **Schmauser Stiftung**, *FAU Erlangen-Nuremberg, Germany*, travel support (6000 Euros)

2009–2010 **CMUC Post-doc scholarship**, *Universidade de Coimbra, Portugal*

2006–2009 **Scholarship of the German Research Foundation (DFG)**, *University of Leipzig, Germany*, Junior research team "Analysis, geometry and their interaction with the natural sciences", PhD

2003–2004 **Study abroad scholarship**, *Australia/ New Zealand*

Awards

2026 **Teaching Award of the Faculty of Science**, *FAU Erlangen-Nuremberg, Germany*

2024 **Award for an outstanding Master's thesis**, *received by my student Nick Schneider*

2022 **Performance bonus**, *FAU Erlangen-Nuremberg, Germany*

2019 **Call for W2-professorship "Harmonic Analysis"**, *TU Chemnitz, Germany*, declined

2019 **Women's Prize**, *Faculty of Sciences, FAU Erlangen-Nuremberg, Germany*
1/2 position for a PhD student for 1 year

10/2007 **Exam award**, *FSU Jena, Germany*

Languages

German	Mother tongue	
English	C1	*Good negotiating skills*
Portuguese	B2	*Fluent*
French	B1	*Intermediate*
Spanish	B1	*Intermediate*
Latin	Qualification	

Academic self-administration and Outreach

04/2026- **Master's Admission Committee for Data Science**, *FAU Erlangen*

2025- **Deputy head for the planning of engineering lectures**, *FAU Erlangen*

2024 **Organisation of the international conference Function Spaces, Differential Operators and Nonlinear Analysis (FSDONA-2024)**, *September 2024*, approx. 80 participants
Together with D. D. Haroske, M. Weimar, and G. Byrenheid.

2022 **Admission Master Data Science**, *FAU Erlangen*

2021 **Organization of the orientation week**, *FAU Erlangen*
For 1st year students, together with Prof. C. Meusburger.

2021-2025 **Representative for mid-level academic staff in the math department committee**, *FAU Erlangen*

2020 **Mentor in the ARIADNEtechnat program**, *FAU Erlangen*

2019 **100th anniversary year of Emmy Noether's Habilitation**, *FAU Erlangen*
Presentation.

2019-2021 **Girls and technology internship**, *FAU Erlangen*
Organisation and representation of the section *'How mathematics helps to look inside a human body'.*

2014-2021 **Member of the study commission Energy engineering**, *FAU Erlangen*

2010-2013 **Maintenance of homepage for the Chair of Scientific Computing**, *FAU Erlangen*

2010-2013 **Women's representative**, *FAU Erlangen*
Organization of the Girl's days 2011–2013; voting member in various appointment committees.

Reviewer (selection)

Analysis and Applications
Nonlinear Analysis
Mathematische Nachrichten
Fourier Analysis and Applications
Zeitschrift für Analysis und ihre Anwendungen
Archiv der Mathematik
Journal of Geometric Analysis

References

Available upon request.

Publication List

Schneider, Cornelia, *Google Scholar Profile*
https://scholar.google.de/citations?user=gXnBGWkAAAAJ&hl=de

Book

C. Schneider, *Beyond Sobolev and Besov: Regularity of solutions of PDEs and their traces in function spaces*, Lecture Notes in Mathematics 2291, Springer, 2021, https://doi.org/10.1007/978-3-030-75139-5

Submitted

[43] **J. Billner and C. Schneider**, *Schauder and Riesz bases for higher-dimensional Lebesgue spaces inspired by ReLU neural networks*, submitted, 2026, arXiv:2610.14266

[42] **S. D. Moura and C. Schneider**, *Fractional gradients and nonlocal PDEs in variable exponent spaces*, preprint, 2026, arXiv:2610.08253

[41] **P. Morin, C. Schneider, and N. Schneider**, *Approximation classes for the anisotropic space-time finite element method. An almost characterization*, submitted, 2026, arXiv:2602.14921

Publications

[40] **V. Kulbatov, J. Lang, C. Schneider, and J. Vybíral**, *Bases of Lebesgue spaces formed by neural networks*, to appear in JFAA, 2026, arXiv:2511.23179

[39] **M. Ullrich, C. Schneider, and J. Vybíral**, *Nonlocal techniques for the analysis of deep ReLU neural network approximations*, J. Mach. Learn. Res., 27(11):1–41, 2026, arXiv:2504.04847

[38] **P. Morin, C. Schneider, and N. Schneider**, *Anisotropic approximation on space-time domains*, J. Approx. Theory, 317:106282, 2026, arXiv:2506.19517

[37] **S. Dahlke, M. Hansen, and C. Schneider**, *Weighted Sobolev and Besov regularity of non-linear parabolic PDEs on non-convex polyhedral domains*, Z. Angew. Math. Mech., 106(8),e70538, 2026, arXiv:2105.13355

[36] **M. Hansen and C. Schneider**, *Refined localization spaces, Kondratiev spaces with fractional smoothness and extension operators*, to appear in Z. Anal. Anwendungen, 2024, arXiv:2405.06316

[35] **S. Probst and C. Schneider**, *Improved universal approximation with neural networks studied via affine-invariant subspaces of $L_2(\mathbb{R}^n)$*, Electron. J. Appl. Math. 3(1), 2025, https://doi.org/10.61383/ejam.20253188

[34] **P. Cioica-Licht, C. Schneider, and M. Weimar**, *Sobolev spaces with mixed weights and the Poisson equation on angular domains*, Ann. Scuola Normale superiore - Classe di Scienze, 49, 2025, arXiv:2409.18615

[33] **M. Actis, F. Gaspoz, P. Morin, C. Schneider, and N. Schneider**, *Direct estimates for adaptive time-stepping finite element methods*, J. Complex. 87, 101918, 2025, https://doi.org/10.1016/j.jco.2024.101918

[32] **M. Hansen, B. Scharf, and C. Schneider**, *Relations between Kondratiev and refined localization Triebel-Lizorkin spaces*, J. Approx. Theory 106162, 2025, https://doi.org/10.1016/j.jat.2025.106162

[31] **M. Hansen and C. Schneider**, *Embeddings and pointwise multiplication in Kondratiev spaces on polyhedral type domains*, Z. Anal. Anwendungen, 44(1/2): 35–77, 2025, https://doi.org/10.4171/ZAA/1779

[30] **C. Schneider and J. Vybíral**, *A multivariate Riesz basis of ReLU neural networks*, Appl. Comput. Harmon. Anal., 68: 101605, 2024, https://doi.org/10.1016/j.acha.2023.101605

[29] **H. Kempka, C. Schneider, and J. Vybíral**, *Path regularity of Brownian motion and Brownian sheet*, Constr. Approx., 2023, https://doi.org/10.1007/s00365-023-09647-z

[28] **C. Schneider and F. Szemenyei**, *Besov regularity of inhomogeneous parabolic PDEs* , Partial Differ. Equ. Appl., 4(43):1–61, 2023, https://doi.org/10.1007/s42985-023-00262-y

[27] **M. Actis, P. Morin, and C. Schneider**, *Approximation classes for adaptive time-stepping finite element methods*, IMA J. Numer. Analysis 43(5):2817–2855, 2023, https://doi.org/10.1093/imanum/drac056

[26] **S. Dahlke and C. Schneider**, *Anisotropic Besov Regularity of parabolic PDEs*, Pure Appl. Funct. Anal. 8(2): 457–476, 2023, arXiv:2112.09485

[25] **F. Szemenyei and C. Schneider**, *Sobolev meets Besov: Regularity for the Poisson equation with Dirichlet, Neumann and mixed boundary values*, Anal. and Appl., 20(5):989–1023, 2022, https://doi.org/10.1142/S0219530522500026

[24] **M. Hansen, F. Szemenyei, and C. Schneider**, *An extension operator for Sobolev spaces with mixed weights*, Math. Nachr. 295(10):1969–1989, 2022, https://doi.org/10.1002/mana.202000590

[23] **S. Dahlke and C. Schneider**, *Regularity in Sobolev and Besov spaces for parabolic problems on domains of polyhedral type*, J. Geom. Anal. 31(12):11741–11779, 2021, https://doi.org/10.1007/s12220-021-00700-6

[22] **D. D. Haroske, C. Schneider, and K. Szarvas**, *Growth envelopes of some variable and mixed function spaces*, J. Geom. Anal. 32(3), paper no. 94, 2022, https://doi.org/10.1007/s12220-021-00811-0

[21] **S. Dahlke, M. Hansen, C. Schneider, and W. Sickel**, *On Besov regularity of solutions to nonlinear elliptic partial differential equations*, Nonlinear Anal., 192, 111686, 2020, https://doi.org/10.1016/j.na.2019.111686

[20] **S. Dahlke, Q. Jahan, C. Schneider, G. Steidl, and G. Teschke**, *Traces of shearlet coorbit spaces on domains*, Applied Math. Letters, 91:35–40, 2019, https://doi.org/10.1016/j.aml.2018.11.019

[19] **S. D. Moura, J. S. Neves, and C. Schneider**, *Traces and extensions of generalized Besov-Morrey and Triebel-Lizorkin-Morrey spaces on domains*, Nonlinear Anal., 181:311–339, 2019, https://doi.org/10.1016/j.na.2019.01.003

[18] **N. Große and C. Schneider**, *Symmetries on manifolds: Generalizations of the radial lemma of Strauss*, Rev. Mat. Complut., 32:365–393, 2019, arXiv:1803.05351

[17] **S. Dahlke and C. Schneider**, *Besov regularity of parabolic and hyperbolic PDEs*, Anal. Appl., 17:235–291, 2019, https://doi.org/10.1142/S0219530518500306

[16] **S. Dahlke and C. Schneider**, *Describing the singular behaviour of parabolic equations on cones in fractional Sobolev spaces*, GEM, 9:293–315, 2018, https://doi.org/10.1007/s13137-018-0106-2

[15] **D. D. Haroske, C. Schneider, and L. Skrzypczak**, *Morrey spaces on domains: Different approaches and growth envelopes*, J. Geom. Anal., 28:817–841, 2018, https://doi.org/10.1007/s12220-017-9843-y

[14] **D. D. Haroske, S. D. Moura, C. Schneider and L. Skrzypczak**, *Unboundedness properties of smoothness Morrey spaces of regular distributions*, Sci. China Math., 60(12): 2349–2376, 2017, https://doi.org/10.1007/s11425-017-9113-9

[13] **S. D. Moura, J. S. Neves, and C. Schneider**, *Spaces of generalized smoothness in the critical case: Optimal embeddings, continuity envelopes, and approximation numbers*, J. Approx. Theory, 187: 82-117, 2014, https://doi.org/10.1016/j.jat.2014.07.010

[12] **N. Große and C. Schneider**, *Sobolev spaces on Riemannian manifolds with bounded geometry: general coordinates and traces*, Math. Nachr., 286(16):1586-1613, 2013, https://doi.org/10.1002/mana.201300007

[11] **S. D. Moura, J. S. Neves, and C. Schneider**, *On trace spaces of 2-microlocal Besov spaces with variable integrability*, Math. Nachr., 286(11-12):1240-1254, 2013, https://doi.org/10.1002/mana.201200092

[10] **C. Schneider and J. Vybíral**, *Non-smooth atomic decompositions, traces on Lipschitz domains, and pointwise multipliers in function spaces*, J. Funct. Anal., 264(5):1197-1237, 2013, https://doi.org/10.1016/j.jfa.2012.12.005

[9] **C. Schneider and J. Vybíral**, *Homogeneity property of Besov and Triebel-Lizorkin spaces*, J. Funct. Spaces Appl., S. Art. ID 281085, 17pp., 2012, https://doi.org/10.1155/2012/281085

[8] **S. D. Moura, J. S. Neves, and C. Schneider**, *Optimal embeddings of spaces of generalized smoothness in the critical case*, J. Fourier Anal. Appl., 17(5):777-800, 2011, https://doi.org/10.1007/s00041-010-9155-0

[7] **C. Schneider**, *Trace operators on fractals, entropy and approximation numbers*, Georgian Math. J., 18(3):549-575, 2011, https://doi.org/10.1515/gmj.2011.0030

[6] **C. Schneider**, *Traces in Besov and Triebel-Lizorkin spaces on domains*, Math. Nachr., 284(5-6):572-586, 2011, https://doi.org/10.1002/mana.201010052

[5] **C. Schneider**, *Trace operators in Besov and Triebel–Lizorkin spaces*, Z. Anal. Anwendungen, 29(3):275–302, 2010, https://doi.org/10.4171/ZAA/1409

[4] **C. Schneider and J. Vybíral**, *On Dilation Operators in Triebel–Lizorkin spaces*, Funct. et Appr., 41(2):139-162, 2009, https://doi.org/10.7169/facm/1261157806

[3] **C. Schneider**, *Spaces of Sobolev type with positive smoothness on $\mathbb{R}^n$, embeddings and growth envelopes*, J. Funct. Spaces Appl., 7(3):251–288, 2009, https://doi.org/10.1155/2009/815676

[2] **D. D. Haroske and C. Schneider**, *Besov spaces with positive smoothness on $\mathbb{R}^n$, embeddings and growth envelopes*, J. Approx. Theory, 161(2): 723-747, 2009, https://doi.org/10.1016/j.jat.2008.12.004

[1] **C. Schneider**, *On Dilation Operators in Besov spaces*, Rev. Mat. Complut., 22(1):111–128, 2009, https://doi.org/10.5209/rev_REMA.2009.v22.n1.16324

Theses

[III] **C. Schneider**, *Besov regularity of partial differential equations, and traces in function spaces*, Habilitation thesis, Philipps-University Marburg, Germany, 2020

[II] **C. Schneider**, *Besov spaces with positive smoothness*, Dissertation, Leipzig University, Germany, 2009

[I] **C. Schneider**, *Growth envelopes in Sobolev spaces*, Diplomarbeit (Master Thesis), FSU Jena, Germany, 2006

Teaching (@ FAU Erlangen)

WS 26/27	**Lecture**, *Methods for Mathematical Data Analysis*
	Lecture, *Higher Analysis (Functional Analysis and Measure Theory for Data Science)*
SS 2026	**Lecture**, *Neural network theory*
	Lecture, *Approximation Theory*
WS 25/26	**Lecture**, *Higher Analysis (Functional Analysis and Measure Theory for Data Science)*
	Seminar, *Neural Network Approximation*
SS 2025	**Lecture**, *Mathematics for Engineers A2*
	Seminar, *Neural Network Approximation*
WS 24/25	**Lecture**, *Mathematics for Engineers A1*
SS 2024	**Lecture**, *Approximation theory*
	Seminar, *Neural Network Approximation*
SS 2023	**Lecture**, *Mathematics for Engineers E2*
WS 22/23	**Lecture**, *Analysis III*
	Lecture, *Mathematics for Engineers E1*
SS 2022	**Lecture**, *Analysis II*
WS 21/22	**Lecture**, *Analysis I*
SS 2021	**Seminar**, *Partial differential equations*, online course
WS 20/21	**Lecture**, *Partial differential equations*, online course
SS 2020	**Lecture**, *Distributions, Sobolev spaces and elliptic differential equations*, online course
WS 2019/20	**Lecture**, *Mathematics for Engineers A3*
	Lecture, *Regularity theory of elliptic PDEs*
SS 2019	**Lecture**, *Mathematics for Engineers A2*
WS 18/19	**Seminar**, *Function spaces*
WS 16/17	**Lecture**, *Mathematics for Engineers A1*
SS 2016	**Lecture**, *Mathematics for Engineers A4*
WS 15/16	**Lecture**, *Mathematics for Engineers A3*
SS 2015	**Lecture**, *Mathematics for Engineers A2*
WS 14/15	**Lecture**, *Mathematics for Engineers A1*
	Seminar, *Interpolation spaces and Applications in Numerical Analysis*
SS 2013	**Lecture**, *Mathematics for Engineers A2*
WS 12/13	**Lecture**, *Mathematics for Engineers A1*
SS 2012	**Lecture**, *Mathematics for Engineers A4*
WS 11/12	**Lecture**, *Mathematics for Engineers A3*
SS 2011	**Lecture**, *Mathematics for Engineers A2*
WS 10/11	**Lecture**, *Mathematics for Engineers A1*

Previous teaching

SS 2009 **Teaching assistant**, *Complex Analysis*, University of Leipzig, Germany

WS 08/09 **Teaching assistant**, *Analysis I for Physicists*, FSU Jena, Germany

WS 08/09 **Teaching assistant**, *Measure and Integral Theory*, University of Leipzig, Germany

Supervised students - PhD

since 10/2026 **Julian Billner**, *Dissertation*, Overcoming the curse of dimensionality: A basis-theoretic approach to neural network approximation (working title), FAU Erlangen-Nuremberg, 20XX

since 01/2024 **Nick Schneider**, *Dissertation*, Adaptive methods for solving evolution equations on non-smooth domains: Convergence rates and approximation classes studied via regularity in anisotropic Besov spaces (working title), FAU Erlangen-Nuremberg, 20XX

2019-2022 **Florá Orsolya Szemenyei**, *Dissertation*, Besov regularity of elliptic and parabolic PDEs with inhomogeneous boundary conditions on Lipschitz domains, FAU Erlangen-Nuremberg, 2022
now: Universa Versicherung, Nuremberg

Supervised students - Master thesis

Duygu Nacar, *Nonlinear approximation rates and Besov regularity for elliptic PDEs on polyhedral domains*, (current)

Felix Dauerer, *Path regularity of Brownian motion*, (current)

Jonas Krüger, *Approximation with neural networks in variable Lebesgue spaces*, 2026

Julian Billner, *Schauder and Riesz bases formed by inner products of higher dimensional Lebesgue spaces inspired by ReLU neural networks*, 2026

Priyanka Yadav, *Nonlinear approximation and deep ReLU networks: ReLU networks are at least as expressive as free-knot linear splines*, 2025
From Data Science.

Nick Schneider, *Approximation classes for adaptive time-stepping finite element methods*, 2023
Received the Fritz and Maria Hofmann Prize for his excellent Master's thesis.

Metin Bozkurt, *Sobolev regularity of parabolic PDEs with inhomogeneous boundary data*, 2022

Florá Orsolya Szemenyei, *General Lipschitz spaces and Sobolev's embedding theorem*, 2019

Supervised students - Bachelor thesis

Henry Huhnt, *Reproducing Kernel Hilbert spaces and applications to machine learning*, (current)
From Data Science.

Sofia Schlicht, *Approximation of the square function using ReLU neural networks*, (current)
From Data Science.

Josef Angermeier, *Theorems of Stone-Weierstraß and Müntz*, (current)

Julian Billner, *Properties of $C(S, \mathbb{R})$ and applications to neural networks*, 2024

Samuel Probst, *Wiener's Tauberian Theorems*, 2024

Ba Duc Duong, *Applications of Morrey and Campanato spaces to partial differential equations*, 2023

Michael Koch, *Approximation theorem of Weierstraß*, 2023
Bachelor thesis and Staatsexamensarbeit

Thorsten Beischer, *Theorems of Jackson-Bernstein type*, 2022

Kilian Seib, *Generalizations of the approximation theorem of Weierstraß*, 2022

Steven Kellner, *Eigenvalue estimates via entropy and approximation numbers*, 2021

Jinxuan Cheng, *On an extreme class of real interpolation spaces*, 2016